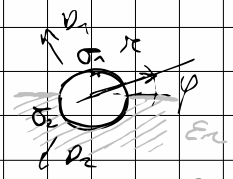
APPELLO 28 SETTEMBRE 2016

ES. 1:

SFERA R; Q PER METÀ IN ε_r



a) $r < R$; $\forall \varphi$: $E = 0$ ESSENDO LA SFERA CONDUTTRICE.

$r > R$, $\varphi \in [0, \pi]$:

$$E(r) = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad \vec{E} = E(r)\vec{u}_r$$

ESSENDO $\nabla \times \vec{E} = 0 \Rightarrow [E_r] = 0$

$r > R$; $\varphi \in [\pi, 2\pi]$ $E(r) = \dfrac{Q}{4\pi\varepsilon_0 r^2}$; $\vec{E} = E(r)\vec{u}_r$

b) $\vec{D}_1 = \varepsilon_0 \vec{E}$; $\vec{D}_2 = \varepsilon_0 \varepsilon_r \vec{E} = \varepsilon \vec{E}$

$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} \Rightarrow \vec{P}_1 = 0$, $\vec{P}_2 = \varepsilon_0(\varepsilon_r - 1)\vec{E}$

$$\vec{P} \cdot \vec{n} = \sigma_{pol} = (\varepsilon_r - 1)\frac{Q}{4\pi R^2}$$

$$\sigma_1 = \frac{Q}{4\pi R^2} \qquad \sigma_2^{netta} = \sigma_{lib} + \sigma_{pol} = \frac{\varepsilon_r Q}{4\pi R^2}$$

ES. 2:

$L = 2m \qquad R_1 = 10cm \qquad R_2 = 20cm \ (2R_1) \qquad R_1, R_2 \ll L$

$N_1 = 1500 \qquad N_2 = 500$

$I = 20A$

a) $r_1 = 1,5R_1 \ ; \ r_2 = 1,5R_2$

ESSENDO $R_1 < r_1 \Rightarrow B_1(r_1) = 0$

$$B_2(r_1) = \mu_0 I \cdot \frac{N_2}{L} = 6,28 \ mT$$

ESSENDO $R_2 < r_2 \Rightarrow B_1(R_2) = B_2(r_2) = 0$

b) $u_m = \frac{1}{2\mu_0} B^2 \ ; \qquad U_m = \int_\tau u_m d\tau$

$$U_m = \frac{1}{2\mu_0} \int_0^{2\pi} \int_0^L \int_0^R B^2 \, d\theta \, dz \, r \, dr =$$

$$= \frac{\pi L}{\mu_0} \left[\int_0^{R_1} (B_1 + B_2)^2 r \, dr + \int_{R_1}^{R_2} B_2^2 r \, dr \right] =$$

$$= \frac{\pi L}{\mu_0} \frac{R_1^2}{2} \left[\mu_0 I \frac{N_2}{L} - \mu_0 I \frac{N_1}{L} \right]^2 + \frac{\pi L}{\mu_0} \left(\mu_0 I \frac{N_2}{L} \right)^2 \left(\frac{R_2^2}{2} - \frac{R_1^2}{2} \right)$$

$$= \frac{\pi \mu_0}{L} I^2 \frac{R_1^2}{2} (N_2 - N_1)^2 + \frac{\pi \mu_0}{L} I^2 \frac{}{2} N_2^2 (R_2^2 - R_1^2)$$

$$= \frac{\pi \mu_0}{2L} I^2 \left[\cancel{R_1^2 N_2^2} + R_1^2 N_1^2 - R_1^2 N_2 N_1 + N_2^2 R_2^2 - \cancel{N_2^2 R_1^2} \right]$$

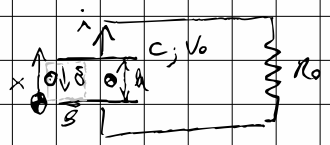
$$= \frac{\pi \mu_0}{2L} I^2 \left[N_2^2 R_2^2 + R_1^2 N_1^2 - R_1^2 N_2 N_1 \right] = 9,8 \ mJ$$

ES. 3:

a) $$\int_\tau \frac{\partial u_{em}}{\partial t}\, d\tau + \int_\tau \vec{E}\cdot\vec{j}\, d\tau + \oiint_{\partial\tau} \vec{P}\cdot\vec{n}\, dS = 0$$

VAR. DI ENERGIA ELETTROMAGNETICA NEL TEMPO, POTENZA DISSIPATA E FLUSSO DEL VETT. DI POYNTING SONO STRETTAMENTE RELAZIONATI.

COND. PIANO S; h. V_0, R_0 SCARICA



$$\vec{P} = \frac{\vec{E}}{\mu_0}\times\vec{B}$$

$$i(t) = \frac{V(t)}{R_0} \qquad Q(t) = C\cdot V(t)$$

$$-\frac{dQ(t)}{dt} = \frac{V(t)}{R_0} = \frac{Q(t)}{CR_0}$$

b) $$\frac{dQ}{Q} = -\frac{dt}{CR_0}$$

$$Q(t) = e^{-\frac{t}{CR_0}}\cdot CV_0 \Rightarrow i(t) = \frac{V_0}{R_0}e^{-\frac{t}{CR_0}}$$

$$\vec{j}_s = \vec{j}_c \qquad \vec{j}_c = \frac{i(t)}{S}\vec{u}_x$$

$$\vec{B} = B(r,t)\,\vec{u}_y = \mu_0\frac{i(t)}{S}\cdot\frac{\pi r^2}{2\pi r} = \frac{\mu_0 i(t)}{2S}r$$

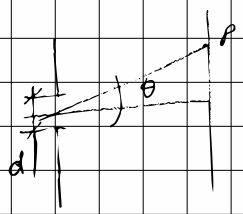
$$\vec{E} = -\frac{V(t)}{d}\vec{u}_x = -\frac{i(t)R_0}{d}$$

$$\vec{P}(r,t) = \frac{i(t)^2 R_0}{2Sd}\cdot r$$

ES. 4:

a) λ ONDA PIANA MONOCROMO, DIFFRAZIONE FENDITURA d.

$$I = I_0 \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2 \quad \text{con} \quad \beta = kd \sin\theta$$



I_0 INTENSITÀ DELL'ONDA PIANA CHE INCIDE SULLA FENDITURA.

b) $d_1 = \lambda$; $d_2 = 10\lambda$

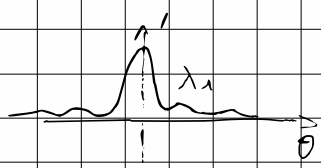
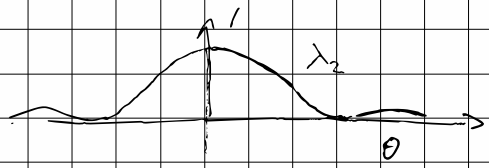
$$\beta_1 = \frac{2\pi}{\cancel{\lambda}} \cancel{\lambda} \sin\theta \qquad \beta_2 = \frac{\cancel{2}\pi}{\cancel{10}\,5} \sin\theta$$

MIN - I ORDINE: $\beta/2 = \pi$

$\Rightarrow \sin\theta_1 = 1 \qquad \sin\theta_2 = 10$

IL MASSIMO PRINCIPALE È CIRCA 10 VOLTE PIÙ LARGO PER λ_2

$$\frac{I_1}{I_0}(0) = \lim_{\theta \to 0} \left(\frac{\sin(\pi \sin\theta)}{\pi \sin\theta} \right)^2 = 1$$

$$\frac{I_2}{I_0}(0) = \lim_{\theta \to 0} \left(\frac{\sin(\frac{\pi}{10} \sin\theta)}{\frac{\pi}{10} \sin\theta} \right)^2 = 1$$

L'ALTEZZA DEI MASSIMI PRINCIPALI COINCIDE.